

SCIENTIFIC QUALITY-CONTROL RECORD

GSE42568 QC Appendix

Input, metadata, processed-expression, annotation, design
and sensitivity review

Dataset: GEO GSE42568 · GPL570

Contrast: cancer minus normal breast tissue

Release: v1.0 · 11 September 2026

Scientific status: PASS WITH DOCUMENTED LIMITATIONS

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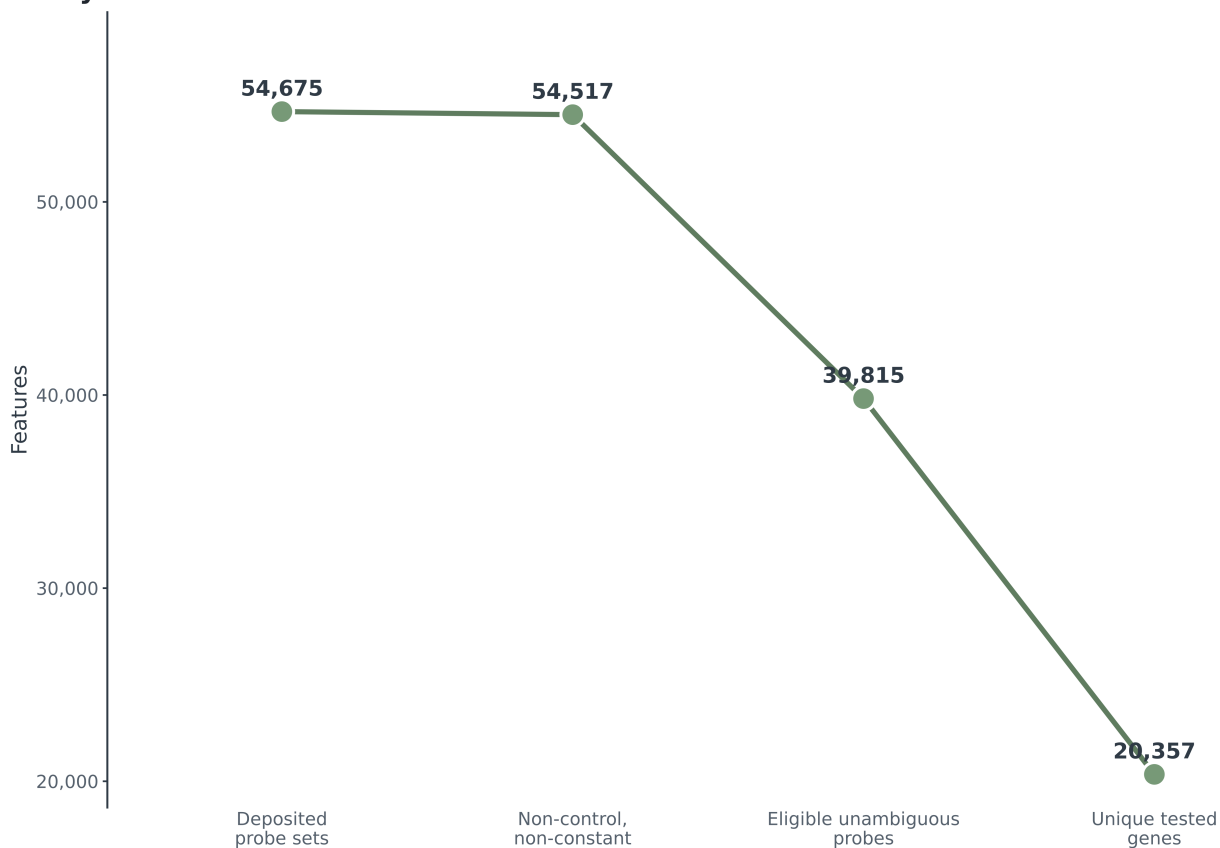
Disposition

PASS WITH DOCUMENTED LIMITATIONS: 32/32 Scientific QA checks and 13/13 reproducibility guards pass; a clean rebuild reproduced 10/10 key tables byte-for-byte. All 121 samples enter P1. Four unusual samples remain after signed review and are removed only in an influence sensitivity.

Input and metadata

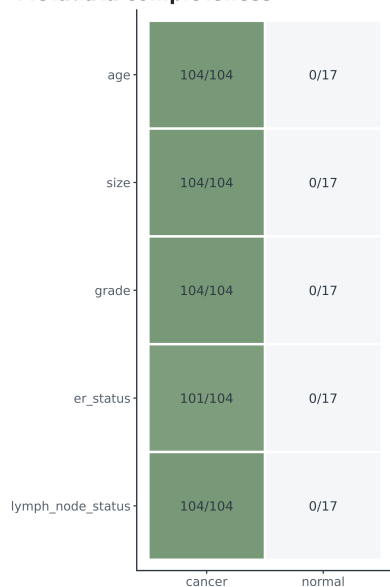
Metric	Observed
matrix_sha256	43b93f72f1d81838dcd2a5b6ccc3bc1875e46230dad0d7346236773a99e32b76
rows_probe_sets	54675
columns_samples	121
unique_probe_ids	54675
unique_sample_ids	121
nonfinite_values	0
minimum	2.3128116420
maximum	16.0469027300
constant_probe_sets	96
AFFX_control_probe_sets	62
cancer_samples	104
normal_samples	17
platform	GPL570
scale_assessment	consistent_with_log2_GCRMA
gzip_integrity	passed

Analysis feature flow

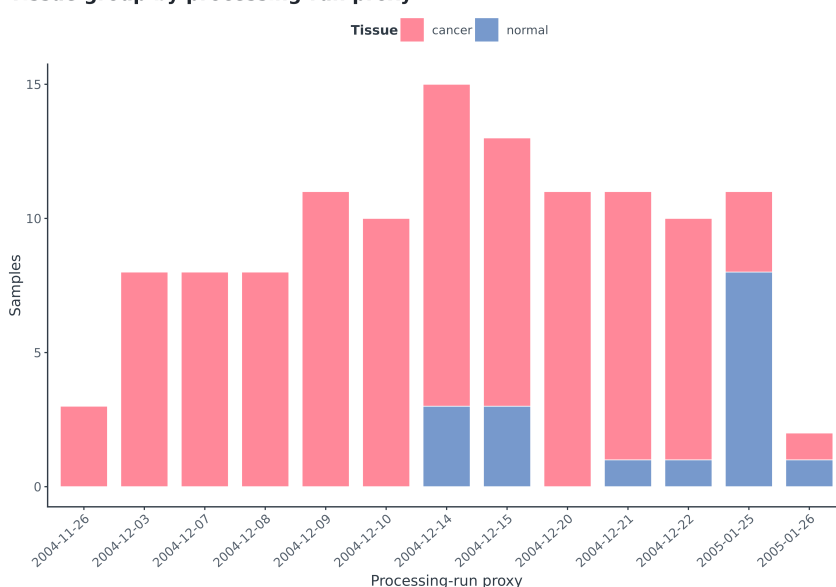


The manifest has 104 cancer and 17 normal GPL570 samples. Normal ages, source category and pairing IDs are absent. GEO's narrative says 20 cancers under 50 while parsed sample fields contain 27 under 50 and 77 at least 50; the discrepancy is preserved, not overwritten.

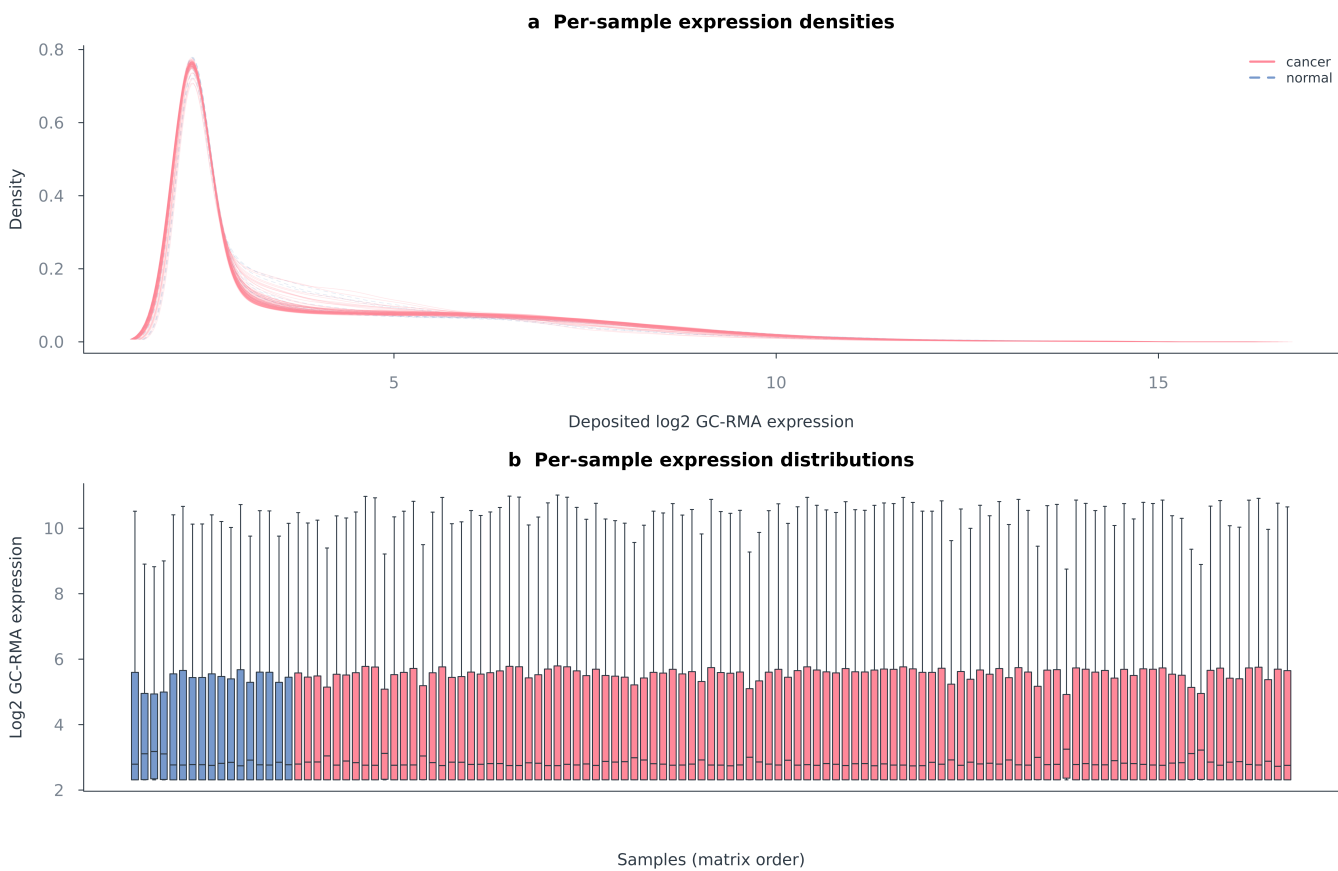
Metadata completeness



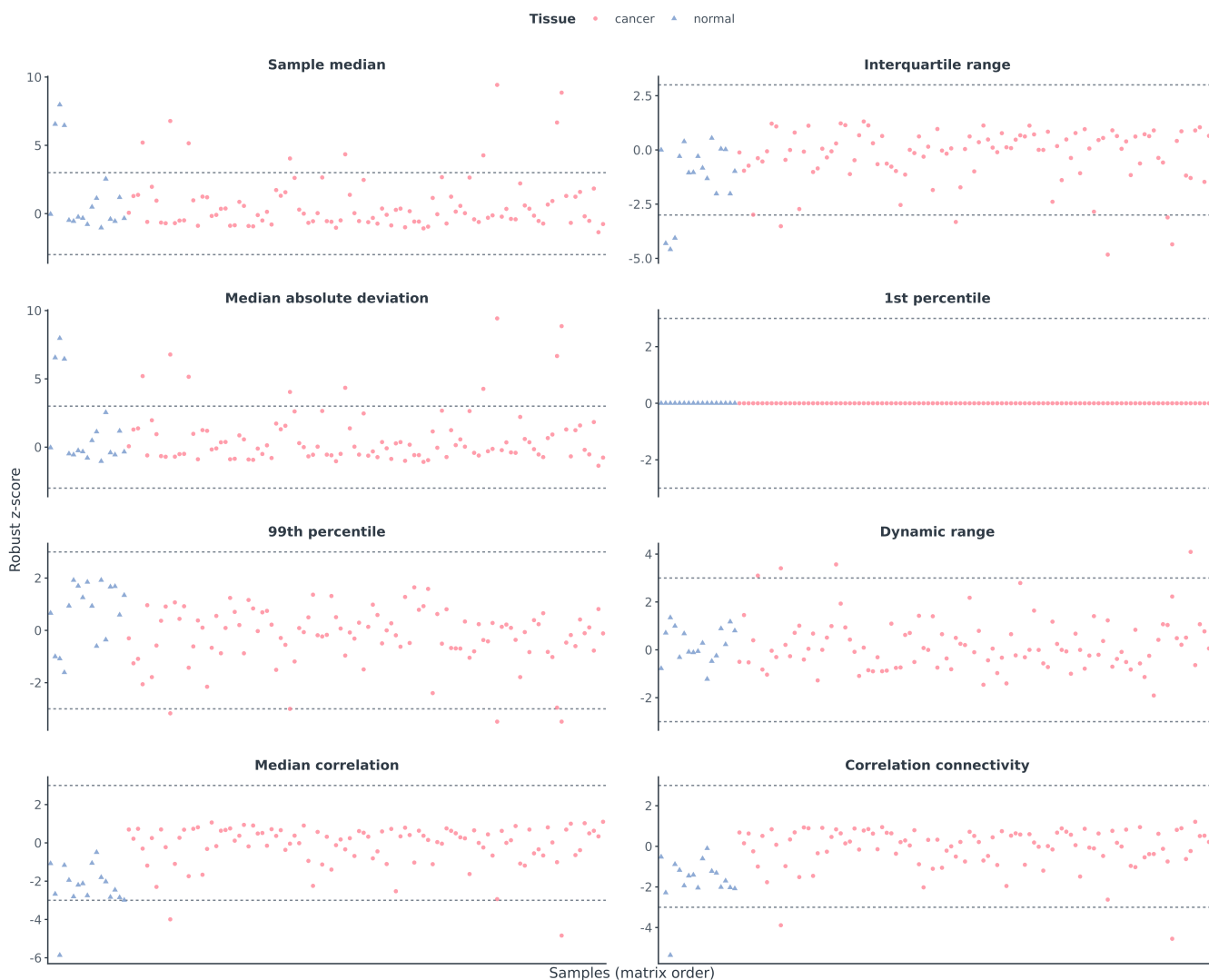
Tissue group by processing-run proxy



Distribution and sample review



Sample-level QC metrics



Sample	Group	Run	Signals	Distribution	Correlation	PCA
GSM1045192	normal	2004-12-14	2	TRUE	FALSE	TRUE
GSM1045193	normal	2004-12-15	3	TRUE	TRUE	TRUE
GSM1045217	cancer	2004-12-09	3	TRUE	TRUE	TRUE
GSM1045302	cancer	2004-11-26	3	TRUE	TRUE	TRUE

The four profiles are unusual but finite and smooth, with plausible expression neighbours. No raw CEL metric corroborates technical failure. They are retained in P1 and jointly excluded only for influence assessment; formal S3 is not applicable.

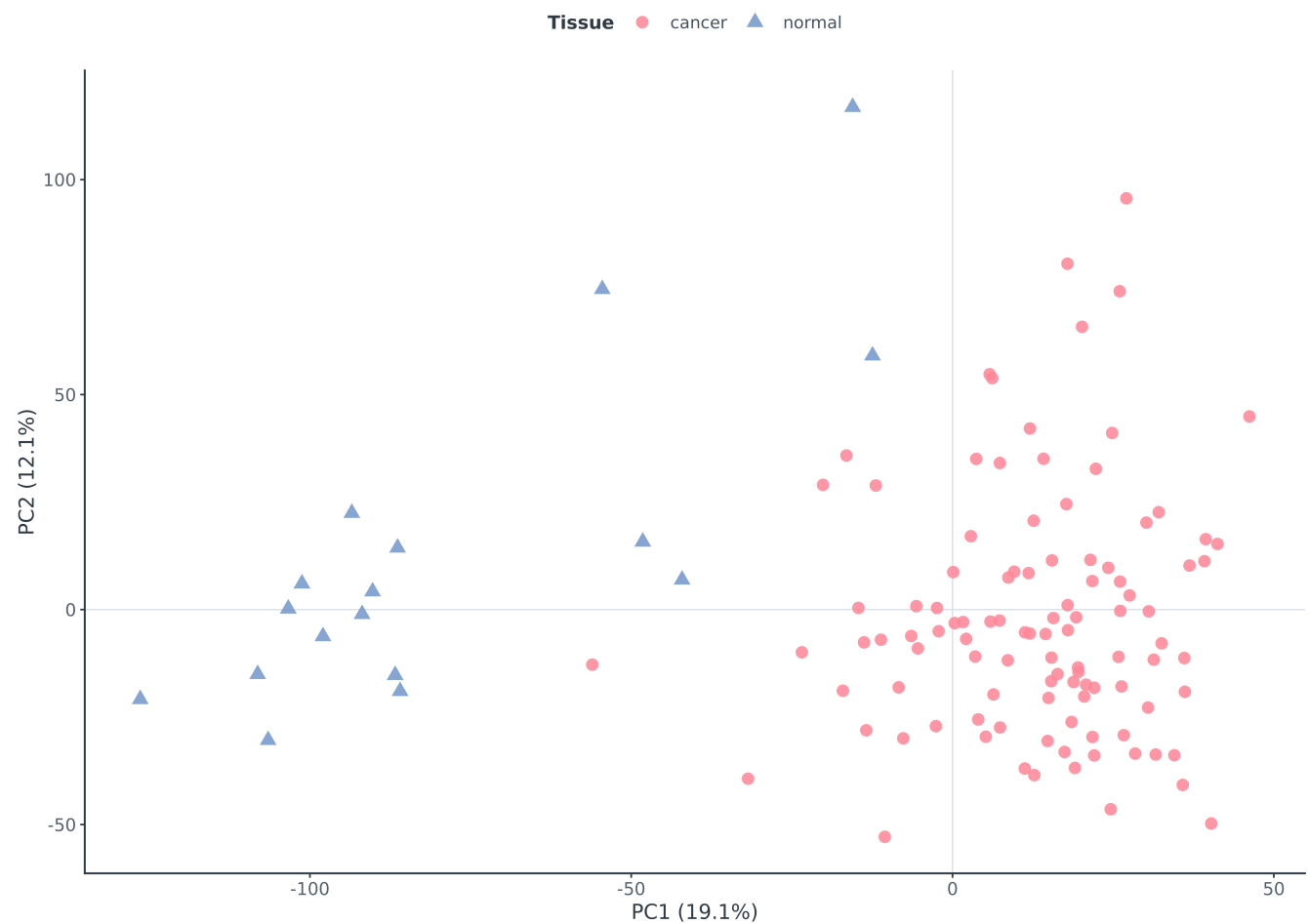
PCA and correlation

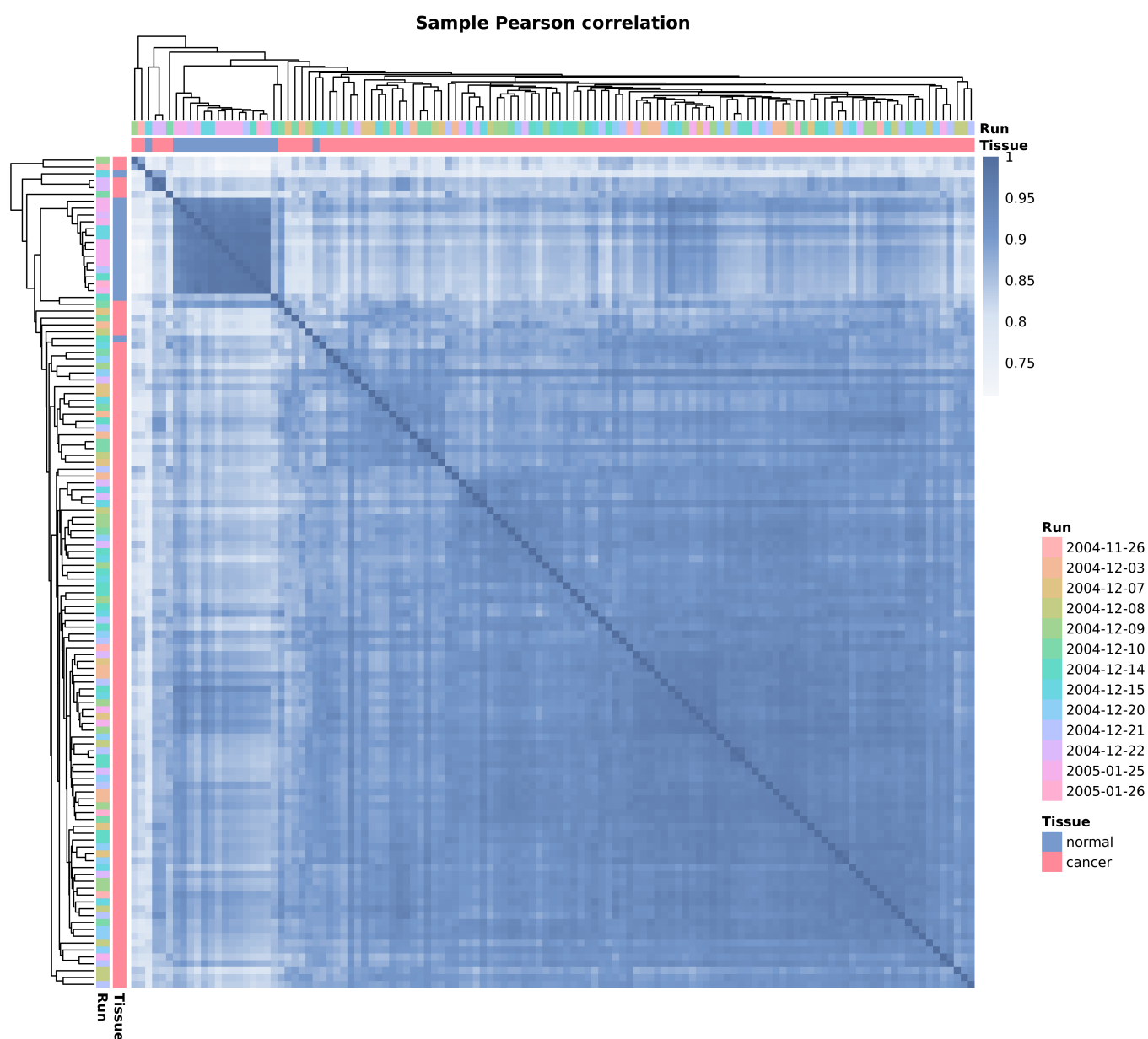
Primary PCA: top 5,000 pooled-MAD genes
Genes centered; not variance-scaled



Visualization-only run-adjusted PCA

Run removed while preserving tissue-group design; never used as the DE response





PC1/PC2 explain 20.3%/10.9% of top-5,000-MAD variation. PC1 tissue R^2 is 0.678; tissue partial R^2 given run is 0.540 and run partial R^2 given tissue is 0.134. Adjusted PCA never enters inference.

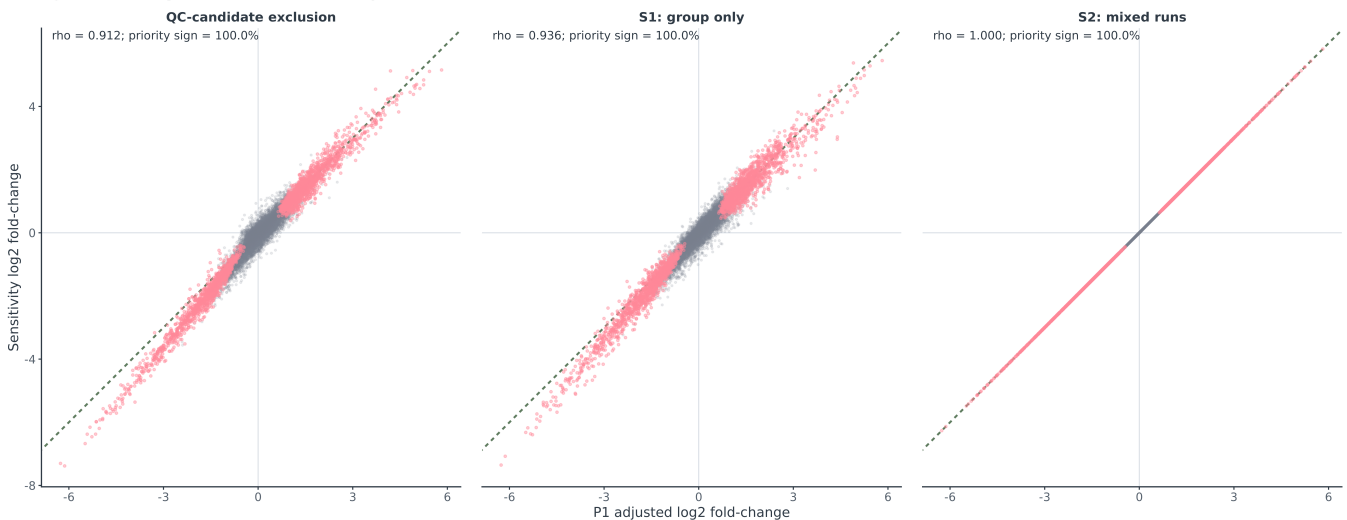
Annotation and statistical QC

Audit metric	Count/value
mapping_status:primary_eligible	39815
mapping_status:unmapped	9953
mapping_status:one_to_many	1524
mapping_status:constant	96
mapping_status:cross_hybridizing_x_at	3225

Audit metric	Count/value
mapping_status:control_affx	62
suffix_class:s_at	11292
suffix_class:at	37394
suffix_class:a_at	1720
suffix_class:x_at	4207
suffix_class:control	62
total_probe_sets	54675
noncontrol_nonconstant_denominator	54517
primary_eligible_probes	39815
primary_eligible_fraction	0.730322651649944
unique_testable_entrez	20357
genes_with_multiple_eligible_probes	10281

Eligibility is 73.03%, above the frozen 60% stop gate; 20,357 genes exceed the 10,000-gene gate. All 54,675 probes receive one exclusive status.

Design and sample-influence stability: Green



Comparison	n	Spearman	Pearson	Priority sign	Median effect ratio
S1_group_only	121	0.936	0.975	100.0%	1.069
S2_mixed_runs	62	1.000	1.000	100.0%	1.000
QC_candidate_exclusion	117	0.912	0.975	100.0%	1.100
S4_no_trend	121	1.000	1.000	100.0%	1.000
S4_group_aware	121	0.962	0.993	100.0%	1.006

Comparison	n	Spearman	Pearson	Priority sign	Median effect ratio
median_collapse	121	0.874	0.902	97.0%	0.960

Removed run	Normal removed	Cancer removed	Spearman	Priority sign
2004-11-26	0	3	1.000	100.0%
2004-12-03	0	8	1.000	100.0%
2004-12-07	0	8	1.000	100.0%
2004-12-08	0	8	1.000	100.0%
2004-12-09	0	11	1.000	100.0%
2004-12-10	0	10	1.000	100.0%
2004-12-14	3	12	0.924	100.0%
2004-12-15	3	10	0.951	100.0%
2004-12-20	0	11	1.000	100.0%
2004-12-21	1	10	0.989	100.0%
2004-12-22	1	9	0.988	100.0%
2005-01-25	8	3	0.953	100.0%
2005-01-26	1	1	0.996	100.0%

The normal-heavy 2005-01-25 removal retains Spearman 0.953 and every priority direction. Median collapse is the least concordant major technical layer (rho 0.874; 97.0% priority direction), and its disagreements are filtered from headline claims rather than hidden.

Enrichment and licence QA

GO BP universe is 15,588; Reactome universe 10,178. GSEA has all 20,357 moderated-t values exactly once and zero exact ties. KEGG and MSigDB Hallmark have explicit expected gate rows; Reactome is the completed fallback.

Acceptance summary

Domain	Checks	Passed	Statuses
DE	4	4	PASS, PASS, PASS, PASS
annotation	4	4	PASS, PASS, PASS, PASS
artifacts	1	1	PASS

Domain	Checks	Passed	Statuses
design	3	3	PASS, PASS, PASS
enrichment	4	4	PASS, PASS, PASS, PASS
input	2	2	PASS, PASS
licensing	2	2	EXPECTED_LICENSE_GATE, EXPECTED_LICENSE_GATE
metadata	5	5	PASS, PASS, PASS, PASS, PASS
sample_QC	2	2	PASS, NOT_APPLICABLE_CONFIRMED_FAILURE_EXCLUSION
scope	1	1	PASS
sensitivity	4	4	PASS, PASS, PASS, PASS

The full row-level ledger is `results/tables/acceptance_checks.tsv`. Residual limitations are normal provenance/age, unpaired bulk tissue, partial run confounding, cancer heterogeneity, processed-only QC and no external validation.